

XRPL Validator — Daily Engineering Report • 2026-09-26

Generated 2026-09-27 00:21 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **4 commits** (xrpl-validator 4)
- Live shadow: 10,531/10,531 ledgers fully matched, 1,373,726 transactions, 0 engine receipts in the current soak
- Amendment **PermissionDelegationV1_1** held majority (since 2026-09-24 21:25 UTC); activation due ~2026-10-08 21:25 UTC
- Amendment **fixBatchV1_2** held majority (since 2026-09-25 14:12 UTC); activation due ~2026-10-09 14:12 UTC
- Amendment **BatchV1_1** held majority (since 2026-09-25 14:46 UTC); activation due ~2026-10-09 14:46 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	10,531
Ledgers fully matched	10,531
Transactions applied natively	1,373,726
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-09-26 12:53 EDT
Reference rippled	3.4.1, full, validated #107,262,909

Engineering changes

- **fix(docs): BatchV1_1 activation is now due ~2026-10-09** (`72e281c` , xrpl-validator) BatchV1_1 lost majority on mainnet for one voting round on 2026-09-25 (ledgers 107,227,905 to 107,228,161) and regained it at 14:46 UTC, which restarted its two-week period: activation is now due about 2026-10-09 14:46 UTC, shortly after fixBatchV1_2 (~2026-10-09 14:12 UTC), instead of ~2026-09-29. PermissionDelegationV1_1 is due about 2026-10-08 21:25 UTC. Comments and the README only.
- **fix(docs): PermissionDelegationV1_1 activation is now due ~2026-10-08** (`c02c3b3` , xrpl-validator) Its mainnet majority restarted on 2026-09-24 (21:25 UTC), so activation is due about 2026-10-08 21:25 UTC rather than ~10-05. Comments only.

- **fix(xrpl-node): replace the dead relay hub; back off and log failed handshakes** (c79ee89 , xrpl-validator) xrplcluster.com has no peer port (the name leads to a web proxy): the validation relay retried it every ~25 s and never connected. The relay hubs move to relay::RELAY_HUBS as slots. Real handshakes on 2026-09-26: s1, s2 and r.ripple.com connect; hub.xrpl-commons.org answers 400 "Failed to verify session" to the same handshake; hubs.xrpkwait.com and zaphod.alloy.ee answer 503 (full). The fourth slot alternates ...
- **fix(xrpl-node): ws-sync returns to the primary WebSocket once it is healthy** (aefd9cc , xrpl-validator) ws-sync moved to the next WebSocket endpoint when a connect failed, but nothing ever moved it back: reset_ws() had no caller. A fallback stint lasted until the next restart — on 2026-09-25 the reference node's xrpld upgrade left ws-sync on a public WebSocket until a warm restart.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #31: clean.** Soak #31 of build 0c76af1 (F402, the voting-ledger validation order and the vote change, deployed 2026-09-25 10:53 EDT) began at 11:20 EDT on 2026-09-25 (seed ledger 107,228,273) and ended at 11:25 EDT today with no engine receipts. The shadow compared 22,125 ledgers with no byte, result or overlay difference, and the state-hash check matched 22,278 consecutive ledgers (107,228,745 to 107,251,022) with no mismatch. The daily receipt canary, run at 12:40 EDT on 2026-09-25, was detected, so the zero count comes from a live comparison.
- **Deployment (validator connections).** Build c79ee89 replaced the live build at 12:54 EDT after its gate (1,685 suite tests and 744 flow-engine tests passing; the flow-engine run now includes campaign 25's vectors) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.4 seconds and resumed one ledger behind the network. Its two changes concern how the validator connects, not the engine:
- The ledger subscription now returns to the reference node on its own. When the reference node cannot be reached the validator moves to a public server, and until now it stayed there until the next restart (on 2026-09-25 a warm restart moved it back). While on a public server it now checks the reference node once a minute and returns as soon as that node holds the network's latest validated ledger.
- The validation relay replaced its fourth hub, which no longer accepts peer connections. Of the hubs tested, three of Ripple's accept our handshake; two community hubs answered that they were full, and one rejected the handshake. The fourth relay now alternates between the two full hubs, and after the deployment it was accepted on its third attempt. A failed attempt is logged with its reason and retried after 10 seconds, doubling to at most 10 minutes, instead of every 25 seconds.

The health check matched 405 ledgers with no mismatches, and soak #32 began at 13:21 EDT (seed ledger 107,252,372). - **Reference node stall and RPC failover.** At 03:16 EDT the reference node stopped answering requests for about four minutes. On the same host, an unrelated maintenance command (a text search across a large directory tree) saturated the system disk, the kernel moved about 5 GB of process memory to swap on that same disk, and the node stalled until the search ended. The validator's RPC client switched to a public fallback server twice (for 10.5 and 316 seconds) and returned to the reference node on its own. Its ledger-by-ledger verification paused after ledger 107,243,458 and resumed with 107,243,459: no ledger was skipped and none mismatched. The engine was not involved. Searches of that kind no longer run on that host. - **Watchdog paging.** The ops copilot re-sent a warning every 15 minutes for as long as it stayed open, so a store-pruning warning that could not clear for days was sent 21 times in five

hours. A warning is now sent once per episode, and again only when a new signal joins it; critical alerts still repeat after a cooldown (copilot commit 4233ece). The warning itself overstated the risk: it projected the fill rate from the last 24 hours, which still held the one-off writes that followed the 2026-09-25 rotation and upgrade (19.6 GB a day). Over the following day the node used about 15 GB a day (10 to 19 GB a day from hour to hour), which leaves about 40 to 50 GB free when the next rotation runs (ledger 107,436,647, about 2026-10-04), against 58 GB at the end of the previous cycle. An earlier version of this note gave 11.6 to 13.2 GB a day and 65 to 75 GB; those figures came from overnight hours only. - **fixBatchV1_2**. Its source is still not published: at 13:17 UTC there was no rippled 3.4.1 tag or release, and neither `develop` nor the 3.4 release and staging branches define the amendment. It keeps majority and activates about 2026-10-09 14:12 UTC, with BatchV1_1 following at about 14:46 UTC.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Sponsored Pull-Charge Authorizations** (fees) Extend the in-flight XLS-67 Charge transaction with a ChargeAuthorization ledger object carrying sfSponsor (AccountID), sfMaxPerCharge, sfMaxTotal, an
- **Size-Proportional Reserves for DID/Credential** (fees) Add sfReserveUnits (uint32) to the DID and Credential ledger objects, computed in doApply from the serialized byte length of DIDSet's Document/URI fie
- **Domain-Scoped Iceberg Offers** (dex) Add a tfIceberg flag to OfferCreate that preflight rejects unless sfDomainID (PermissionedDEX) is also set; when present, the Offer SLE stores sfDispl

Appendix — commits

Time	Repo	Commit	Subject
00:23	xrpl-validator	72e281c	fix(docs): BatchV1_1 activation is now due ~2026-10-09
00:24	xrpl-validator	c02c3b3	fix(docs): PermissionDelegationV1_1 activation is now due ~2026-10-08
12:06	xrpl-validator	c79ee89	fix(xrpl-node): replace the dead relay hub; back off and log failed handshakes
12:06	xrpl-validator	aefd9cc	fix(xrpl-node): ws-sync returns to the primary WebSocket once it is healthy